	University of Rajsahhi
	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
	B.Sc. Engineering Part IV Odd Semester Exam-2022
	Course Code: CSE4111 Course Title: Parallel Processing and Distributed System
Time: 03 Hours	Full Marks: 52.50

SECTION A
ANSWER ANY THREE OF THE FOLLOWINGS

1.(a) Why are we developing parallel systems instead of sequential systems? [3.00]
(b) Why do we need to write parallel programs based on parallel algorithms? [3.25]
(c) How can we exploit instruction level parallelism and loop level parallelism? [2.50]

2.(a) How Flynn classifies computers? Discuss with necessary diagrams. [3.00]
(b) Differentiate between the characteristics of CISC and RISC processor architectures. [2.00]
(c) Briefly explain the architecture, instruction format, and pipeline operations of a typical VLIW processor with necessary diagram. [3.75]

3.(a) Define speedup, efficiency and throughput. [2.25]
(b) Define interleaved memory organization. Compare between high-order memory interleaving and low-order memory interleaving? [3.50]
(c) Draw the interconnection structure of a generalized multiprocessor system. [3.00]

4.(a) Classify pipeline processors according to the levels of processing. [1.75]
(b) Draw and explain the structure of a pipeline floating point adder. [3.50]
(c) Explain the functional segmentation of typical storage-to-register floating point instruction in the IBM 360/91. [3.50]

Property of Seminar Library
Dept. of Computer Science & Engineering
University of Rajshahi.

SECTION B
ANSWER ANY THREE OF THE FOLLOWINGS

5.(a) What is false sharing? Explain with an example. [2.75]
(b) What are the differences among mutex, busy-waiting and semaphore? [3.00]
(c) What is the purpose of using Allgather() function in MPI? [3.00]

6.(a) Distinguish between local and remote distribution. [1.50]
(b) Write the main factors that trend to distributed processing. [3.00]
(c) What are the basic differences between function and system distribution? [2.00]
(d) Write different categories of data distribution. [2.25]

7.(a) Write the role of transaction manager and transaction coordinator for a distributed database. [2.75]
(b) What is fragmentation? Differentiate between horizontal and vertical fragmentation with example. Also write their advantages. [3.50]
(c) What are the main reasons of failure of a distributed transaction? [2.50]

8.(a) Briefly explain two-phase commit protocol. [4.00]
(b) What is replication? Write some advantages and disadvantages of data replication. [2.25]
(c) Define Heterogeneous Distributed Databases. Write its advantages [2.50]

CamScanner

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
University of Rajshahi
B.Sc. Engineering Part IV Odd Semester Examination-2021
Course Code: CSE4111 Course Title: Parallel Processing and Distributed System

Time: 03 Hours Full Marks: 52.50

SECTION A
[ANSWER ANY THREE OF THE FOLLOWINGS]

1.(a) *'Parallel computing is the simultaneous use of multiple computing resources to solve a computational problem'*. Give some real-life examples. Write the main challenges for parallel computing. [3.50]

(b) What are the levels of parallel processing? Explain each of the levels with at least one example. [4.00]

(c) Does parallel processing differ with multiprocessing? Explain your answer. [1.25]

2.(a) What do you mean by **'Flynn's** Taxonomy'? Explain how SIMD and MIMD computers work with instructions and data. [4.50]

(b) *'Parallelism can be achieved in uniprocessor system using some hardware and software technologies'*- what are they? [2.00]

(c) Explain how parallelism can be achieved using ***multiprogramming*** and ***time-sharing*** methods. [2.25]

3.(a) *'Data dependency is a situation in which a program statements refer to the data of a preceding statement'*. What are the types of it? Explain with examples. [4.00]

(b) Consider the following code fragment of a program— [2.75]

S1: Load R1,A
S2: Add R2,R1
S3: Move R1,R3
S4: Store B,R1
Find the dependencies between the statements in the code fragment.

Property of Seminar Library
Dept. of Computer Science & Engineering
University of Rajshahi.

(c) Explain how ***I2*** is resource dependent on ***I1*** in the fragment given below— [2.00]

I1: A=B+C
I2: D=G+H
Hence, draw the dependency graph.

4.(a) Define the following terms: (i) Speedup, (ii) Efficiency, (iii) Throughput. [3.00]

(b) A five-stage linear pipeline has clock period of τ=100ns, and n=50 tasks. What is its throughput, efficiency and speedup? [3.25]

(c) A CPU of modern computer is a good example of linear pipeline. Explain. [2.50]

CamScanner

SECTION B
[ANSWER ANY THREE OF THE FOLLOWINGS]

5.(a) What are the abstract parallel programming models over the hardware and memory architectures? Compare the relative advantages and disadvantages of ***shared memory model*** with ***message passing model***. [4.75]

(b) What do you mean by MPI? Discuss the purposes and uses of the followings— [4.00]
i) MPI_Init, ii) MPI_Finalize, iii) MPI_Send and iv) MPI_Recv

6.(a) What is cache coherency? With a suitable example, explain how cache becomes incoherent due to the ***write-access*** of shared variables from multiple number of processors. [4.00]

(b) When two processors ***P1*** and ***P2*** have same data element ***X*** in their local caches and one processor modifies the value of ***X*** to ***Y***, then the processor's cache becomes inconsistent with the other's processor's cache and the shared memory. Explain this inconsistency problem for the situations of both ***data Sharing*** and ***process Migration,*** in terms of ***write-through*** and ***write-back*** policies. Use suitable examples and figures. [4.75]

7.(a) Suppose you have an array {12, 14, 6, 8, 10, 16, 2, 4}, show all the phases of sorting the array using odd-even transposition sort algorithm. [4.50]

(b) Write a program using OpenMP to implement the odd-even transposition sort algorithm. [4.25]

8.(a) Define linear and non-linear pipelines. Discuss their differences. [3.75]

(b) Consider the following reservation table of a three-stage non-linear pipeline to execute a function **X**. [5.00]

	Time---->							
		1	2	3	4	5	6	7	8
Stage	S1	X					X		X
	S2		X		X				
	S3			X		X		X	

Reservation Table for function **X**

i) What are the forbidden and permissible latencies?

ii) Calculate the collision vector.

iii) Draw the state transition diagram.

CamScanner

Property of Seminar Library
Dept. of Computer Science & Engineering
University of Rajshahi.

University of Rajshahi

Department of Computer Science & Engineering

B.Sc. Engineering Part IV Odd Semester Examination 2020

Course Code: CSE 4111

Course Title: Parallel Processing and Distributed System

Full Marks: 52.5 **Time: 3 Hours**

[Answer any **SIX (06)** questions taking **THREE (03)** from each section]

Section A

1.	a)	Why are we developing parallel systems instead of sequential systems?	3
	b)	Why do we need to write parallel programs?	2.75
	c)	How can we exploit instruction level parallelism and loop level parallelism?	3
2.	a)	If we fetch a cache line from main memory, where in the cache should it be placed?	3.25
	b)	What is the bottleneck of Von Neumann architecture?	1
	c)	Explain which function is **thread safe** and which is not.	2
	d)	What are the limitations of MPI_Scatter and MPI_Gather functions?	2.50
3.	a)	What are the key characteristics of a vector processor?	2.75
	b)	Does false sharing generate incorrect result? Explain the effects of false sharing.	3
	c)	What is cache coherence problem? Write a program to illustrate the cache coherence problem.	3
4.	a)	How can the receiver find out the following values in MPI? 1. The amount of data in the message. 2. The sender of the message. 3. The tag of the message.	3.75
	b)	Why do we need MPI_Reduce and MPI_Allreduce function? Show the communication pattern of MPI_Allreduce.	3
	c)	Write short notes on GPU.	2

Section B

5.	a)	Why do we use MPI_Ssend instead of MPI_Send?	2
	b)	A set of numbers (15, 11, 9, 16, 3, 14, 8, 7, 4, 6, 12, 10, 5, 2, 13, 1) and four processes are given. Show how the processes can sort the numbers using parallel odd-even transposition sort.	3.25
	c)	Write a program in MPI to sort a set of numbers using parallel odd-even transposition sort algorithm.	3.50
6.	a)	Define grain size and latency.	2
	b)	Define different types of data dependence.	3.25
	c)	What are the steps of grain packing? Explain with an example.	3.50
7.	a)	Can multiple threads simultaneously insert new nodes into a linked list without any problem? If not, explain how to insert the nodes and write a code segment to insert the nodes by multiple threads.	3.25
	b)	How can we synchronize threads?	3
	c)	What is mutex? Write a program using mutex.	2.50

CamScanner

8. a) What are the problems of busy-waiting? 2.50

b) Define critical section and race condition. 2.50

c) fibo[0] = fibo[1] = 1; 3.75

```
# pragma omp parallel for num_threads(thread_count)
for (i = 2; i < n; i++)
fibo[i] = fibo[i-1] + fibo[i-2];
```

What will be the results if we try running the above code with more than one thread?

CamScanner

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. (Engg.) Part-4, Odd Semester, Examination-2019

Course: CSE-4111 (Parallel processing and distributed system)

Time: 03 Hours Full Marks: 52.5

Section A

Answer any THREE questions.

1. (a) What do you mean by parallel processing? 1
   (b) Discuss different types of data dependence. 2
   (c) Two 3x3 matrices A and B are given to compute the sum of the nine elements in the resulting product matrix C=AxB. On the other hand, 150 cycles for inter-processor communication latency, 80 cycles for the nodes performing multiplication, 15 cycles for the nodes performing addition are given. Draw a fine-grain program graph. Perform grain packing and parallel scheduling. 3.5
   (d) What is data-flow computer? Write a simple C program and draw its dataflow graph. 2.25

2. (a) How can you define speedup factor, efficiency and throughput of a k-stage pipeline processor over an equivalent non pipelined processor? 2.75
   (b) 5

	1	2	3	4	5	6
S1	X					X
S2		X			X	
S3			X			
S4				X		
S5		X				X

Considering the above reservation table

(a) List the set of forbidden latencies and the collision vector

(b) Draw the state transition diagram

(c) Determine the minimum average latency of this pipeline

   (c) Write some applications of vector processor? 1

3. (a) Define instruction set. Briefly discuss complex instruction set and reduced instruction set. 4.75
   (b) Differentiate among scalar, superscalar, and vector processor. 2
   (c) Which type of memory interleaving can tolerate memory fault and which cannot? Explain. 2

4. (a) What is CUDA kernel function? 1.5
   (b) What are the effects of launching CUDA kernel functions? 2.25
   (c) Two vectors are given. Write a CUDA kernel to perform addition of two vectors. 3
   (d) How can you set grid and thread block dimensions in CUDA programming? 2

CamScanner

Section B

Answer any THREE questions.

5. (a) Define bisection width and bisection bandwidth. 2
   (b) Discuss snooping cache coherence and directory-based cache coherence method. 4
   (c) Write down the Amdahl's law. 2.75

6. (a) Write a program to read a vector and distribute it among processes using MPI provided function. 3.25
   (b) Sort the numbers 15, 11, 9, 16, 3, 14, 8, 7, 4, 6, 12, 10, 5, 2, 13, 1 using parallel odd-even transposition sort algorithm. 3.5
   (c) What is the purpose of using Allgather() function in MPI? 2

7. (a) What is false sharing? Explain with an example. 2.25
   (b) What are the differences among mutex, busy-waiting and semaphore? 3
   (c) Can multiple threads simultaneously delete nodes of a linked list without any problem? If not, explain how to delete the nodes and write a code segment to delete the nodes by multiple threads. 3.5

8. (a) Mention the desired features of a parallel language. 2
   (b) Convert the following sequential code to vector code: 2

```
DO 20 I=1, N
        A(I)=B(I-1)
20 20 B(I)=2*B(I)
```

   (c) Describe the following optimization of vector functions: 2.75+2
      (i) Redundant express elimination.
      (ii) Constant folding at compile time

CamScanner

5

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. (Engg.) Part-4, Odd Semester, Examination-2018

Course: CSE4111 (Parallel Processing and Distributed System)

Marks: 52.5 Time: 3:00 Hours

[N.B. Answer any **Six** questions taking **Three** from each section]

Property of Seminar Library
Dept. of Computer Science & Engineering
University of Rajshahi.

Section-A

1. (a) Why are we developing parallel systems instead of sequential systems? 2
   (b) If a fraction r of our serial program remains unparallelized , then Amdal,s law says we can't get a speedup better than 1/r. Illustrate with an example. 3
   (c) Suppose S is a shared variable that is initialized to 3. A0, B0 and C0 are private and owned by core 0, on the other hand A1, B1, C1 and A2 are private and owned by core 1. What will be the value of A2 after executing the following statements? How to solve the problem (if you find any)? 3.75

Time	Core 0	Core 1
0	A0=S;	A1=3*S;
1	S=5;	B1=C1+1;
2	B0=C0+2;	A2=4*S;

2. (a) What happens when multiple processes first call MPI_Send simultaneously and then call MPI_Recv? How can you make your program safe if it is unsafe due to the use of MPI_Send and MPI_Recv? 3
   (b) Explain why we use MPI_Allgather. 2
   (c) Write an MPI program that implements multiplication of a vector by a scalar and dot product. The user should enter two vectors and a scalar, all of which are taken by process 0 and distributed among the processes. The results are calculated and collected from process 0, which prints them. You can assume that n, the order of the vectors, is evenly divisible by comm_sz. 3.75

3. (a) Suppose you have an array {12, 14, 6, 8, 10, 16, 2, 4}, show all the phases of sorting the array using odd-even transposition sort algorithm. 4.75
   (b) Write an OpenMP program to implement the odd-even transposition sort algorithm. 4

4. (a) Discuss the efficiency of pipelined processor over non pipelined processor for a single instruction? 3
   (b) What is pipelining? To perform pipelining in a processor what steps must be followed. 2
   (c) Draw and explain the structure of a linear pipeline for floating-point multiplication. 3.75

CamScanner

Section-B

5. (a) Define software parallelism and hardware parallelism. 1.75
   (b) Is there any relation between software parallelism and hardware parallelism? Explain. 3
   (c) Perform a data dependence analysis on each of the following two program segments. Show the dependence graphs among the statements with justification. 4

(i) S1:	A=B+D	(i) S1:	X=SIN(Y)
(ii) S2:	C=Ax3	(ii)S2:	Z=X+W
(iii)S3:	A=A+C	(iii)S3:	Y= -25xW
(iv)S4:	E=A/2	(iv)S4:	X=COS(Z)

6. (a) How to eliminate communication delays between processors through node duplication? Illustrate with an example. 3
   (b) Suppose two 2x2 matrices A and B are multiplied to compute the sum of the four elements in the resulting product matrix C=AxB. Draw a fine-grain program graph assuming each node requires 101 CPU cycles for multiplication, 8 CPU cycles for addition and 212 CPU cycles for inter-processor communication latency. Perform grain packing and show sequential schedule (before grain packing) and parallel schedule for 4 and 6 processors (after grain packing). 5.75

7. (a) Consider the following reservation table 5

	1	2	3	4	5	6	7	8
S1	X					X		
S2		X		X				X
S3			X		X		X	

   (i) List the set of forbidden latencies and the collision vector
   (ii) Draw a state transition diagram showing all possible initial sequences (cycles) without causing a collision in the pipeline
   (iii)List all the simple cycles from the state diagram
   (b) What is the function of scoreboard in dynamic instruction scheduling? 2.75
   (c) How to calculate throughput of a pipeline? 1

8. (a) Define distributed system? Why distributed systems to be needed? 3
   (b) Differentiate between parallel and distributed systems. 2
   (c) What kind of naming services do you know? Why do distributed systems need this function? 2
   (d) What is distributed deadlock and why are they hard to detect? 1.75

CamScanner